

JavonLoong

Curriculum Vitae • 纪文龙

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🌐 JavonLoong



Profile

Undergraduate student at Xingjian College, Tsinghua University, pursuing a dual-degree track in Theoretical and Applied Mechanics and Energy and Power Engineering. Expected graduation: July 2028. Focus areas include AI application engineering, RAG / GraphRAG, engineering simulation, product prototyping, and 3D parametric modeling.

Education

2024.09–2028.07 **B.Eng. candidate, Tsinghua University, Xingjian College, Beijing, Expected**
Dual-degree track: Theoretical and Applied Mechanics + Energy and Power Engineering.

Skills

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| AI engineering | RAG / GraphRAG exploration, knowledge-base construction, vector search, backend control panels, and engineering validation. |
| Simulation | COMSOL modeling, Python numerical simulation, engineering-result reproduction, and parametric experiments. |
| Product | Requirement analysis, interaction flows, agent prototyping, AIGC 3D generation, and presentation narrative. |
| Workflow | Tool-first research, context engineering, multi-agent collaboration, human review, and staged verification. |

Selected Projects

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| RAG / GraphRAG | Power-equipment RAG/GraphRAG intelligent QA system, Core developer, FastAPI, Chroma, hybrid retrieval <ul style="list-style-type: none">○ Built a local, source-traceable QA system for gas-turbine and wind-equipment materials where manual lookup was fragmented and AI answers needed evidence.○ Owned document parsing/OCR cleanup, chunking, vector ingestion, FastAPI retrieval APIs, GraphRAG graph capability, automated evaluation, and Web console delivery.○ Processed 6,098 pages into 593 auditable chunks, supporting demonstrable, measurable, and evidence-grounded professional QA.○ Demo: javonloong.github.io/RAG; Source: github.com/JavonLoong/RAG. |
| Simulation | High-altitude wind-energy AWES and COMSOL reproduction, COMSOL, Python, numerical validation <ul style="list-style-type: none">○ Reproduced representative flight trajectories, tether tension, power curves, and pumping-cycle energy accounting.○ Connected COMSOL aerodynamic polar data to a Python main simulation and organized a 42-case review loop. |
| AI Agent / AIGC | Smart electronic-product innovation course project, Agent prototype, interaction flow, 3D generation <ul style="list-style-type: none">○ Completed agent concept design, requirement decomposition, interaction-flow design, and functional prototyping on the iFLYTEK Xingchen platform.○ Used Tencent Hunyuan 3D for model generation and refinement; received course awards for Agent design and 3D model design. |
| Visual communication | Aesthetic-education image and presentation production, Image generation, layout control, presentation narrative <ul style="list-style-type: none">○ Produced presentation materials covering asset selection, image style, page redraw, and final delivery.○ Focused on visual judgment, information hierarchy, layout stability, and reviewable deliverables. |

- 3D parametric modeling **MoonCake Studio mooncake-mold designer, *Three.js*, real-time preview, STL / GLB export**
- Implemented a browser-based 3D mooncake-mold customization tool with edge profiles, patterns, Chinese engraving, image relief, and dual-view preview.
 - Supported STL / GLB export, covering 3D modeling, interaction design, and production-oriented file output.
 - Demo: jvonloong.github.io/Mooncake-Modle; Source: github.com/JvonLoong/Mooncake-Modle.

Leadership and Public Service

- 2025.05–Present **Vice President, *Tsinghua University Zhi Shi Zhe Association***
- Helped host Bolivian Ambassador to China Hugo Siles and coordinated a demonstration-cooking event for cross-cultural exchange.
 - Refined event agenda, material checklists, on-site staffing, and member co-creation workflow to stabilize themed cooking and exchange activities.
- Winter 2026 **Team leader / first-cohort fellow, *E-Qi Xiang Weilai “Jiuyuan Qinghe” Baotou teaching and research practice team***
- Conducted field research on Baotou’s rare-earth processing, defense manufacturing, urban ecology, and youth-talent retention under the “Lucheng Little Gear Program”.
 - Delivered STEM outreach at Baotou No. 95 Middle School, translating digital logic, AI, computing, angular momentum, and entropy into accessible sessions.
 - Synthesized policy suggestions on practice-to-internship-to-employment pathways, pilot-scale validation bases, productive-service platforms, and rare-earth standard-setting power.
- 2024–2025 **Volunteer, public speaker, and tutor, *Campus service and teaching***
- Served in admissions volunteering, campus guiding, “Return to Alma Mater” visits, Hongyan training-camp service, and large exhibition volunteering.
 - Tutored students in mathematics, physics, Chinese, and full-subject review, emphasizing clear decomposition of complex material.

Honors and Awards

- Scholarships Tsinghua University Strong Foundation Plan A+ rating; 2024 Haichang Intelligent Scholarship First Prize; 2024 Dushulang Yuanmeng Scholarship; 2025 National Inspirational Scholarship; 2025 Tsinghua University Volunteer Public Welfare Scholarship.
- Competitions Third Prize, Beijing 36th Undergraduate Mathematics Competition; Third Prize, 2025 Chinese Mathematics Competition for College Students, Beijing Division; Third Prize, 2025–2026 Tsinghua Theoretical Mechanics Competition.
- Innovation iFLYTEK Xingchen Agent Design Excellent Creativity Second Prize; Tencent Hunyuan 3D Model Design Excellent Creativity First Prize; Tsinghua Xingjian College “Wuxing Cup” Most Feasible Award.
- Other 2024 undergraduate military-training advanced individual; 2021–2022 Hebi Senior High School outstanding student; 2022 Hebi “Three-Good Student”; 2023 provincial second prizes in mathematics, physics, and biology competitions, and third prize in chemistry competition.